

Scientific Analysis of Moral Judgments using Tobit Models with Four Empathy Constructs

Automated Research Pipeline

abril 11, 2026

1 Dataset and Sample Description

The empirical foundation of this project rests on two primary experimental datasets: FLORIDA and BUC. The sample consists of students from both the Floridablanca and Bucaramanga Campuses. These datasets capture incentivized moral judgments from distinct socio-economic contexts. Participants were presented with standardized negotiation scenarios where a negotiator's decision resulted in varying degrees of payoff for themselves, their own group, and a victim group. Under Option 2: judgment-level relational modeling, each judgment is treated as relational and linked both to a descriptive victim x negotiator scenario shorthand and to hypothesis-specific predictors for N1 (judged negotiator) and N2 (counterpart), observer-side victim alignment when applicable, and additional relational controls.

2 Datacard and Variable Definitions

The following table defines the primary symbols and variables used in the mathematical specifications and hypothesis tests.

Table 1: Symbols and Variable Dictionary.

Symbol	Definition
y_{isjr}	Observed moral judgment (Scale: -9 to 9).
y_{isjr}^*	Latent moral preference score (unbounded true judgment).
β_1	The measured effect of a variable on moral judgment (coefficient).
IRI_i	Empathy score: Participant's propensity for empathy across different IRI subscales.
$\mathbf{S}_{isj}$	Relationship structure: The group status of both N1 and N2 negotiators (Ingroup, Outgroup, or Control).
V_{is}	Observer-victim match: Whether the observer and victim have matching group affiliations.
G_{isjr}	N1 Group Status: The specific group affiliation of the N1 negotiator being judged.
A_{is}	Action Taken: Whether the harm was Accepted (A=1) or Rejected (A=0).
C_{isjr}	Relational Controls: Other factors that might influence judgment, like age or socioeconomic status.
ICC	Intraclass Correlation: The degree of similarity in judgments within the same participant.

Option 2: judgment-level relational modeling retains a victim x judged-negotiator shorthand for descriptive summaries, while the executable hypotheses use explicit negotiator-level structure terms. H2 now uses the judged-plus-counterpart structure directly, and in observer rows it adds the player-victim alignment term and its interaction with that structure.

2.1 Predictor Glossary and Abbreviations

The regression tables and dynamic figures use compact predictor labels to stay readable. The bullet list below maps those compact labels back to their full meaning and subset-specific interpretation.

- `iri_fs`, `iri_ec`, `iri_pt`, `iri_pd` [FS, EC, PT, PD]: Active empathy predictors: fantasy, empathic concern, perspective taking, and personal distress. Used in H1/H2/H3 active model.
- `judged_ingroup`, `judged_outgroup` [N1 In, N1 Out]: N1 contrasts relative to the N1 control-labeled baseline. Used in H1/H3.
- `counterpart_ingroup`, `counterpart_outgroup` [N2 In, N2 Out]: N2 contrasts relative to the N2 control-labeled baseline. Used in H1/H3.
- `decision_accept` [Acc]: Accepted harmful deal relative to rejected harmful deal. Used in H1/H3.
- `observer_victim_outgroup` [V Out (Obs)]: Observer-only victim outgroup contrast; excluded from victim-only formulas. Used in H1/H3 Bystander only.
- `h2_negstruct_*` [N1 ..., N2 ...]: H2 joint N1/N2 structure dummies with reference N1 Ctl, N2 Ctl. Used in H2 Victim and Bystander.
- `player_victim_outgroup` [V Out]: Observer-side player-victim outgroup contrast with player-victim ingroup as the reference. Used in H2 Bystander only.
- `player_victim_outgroup:h2_negstruct_*` [V Out x N1 ..., N2 ...]: Bystander-only H2 interaction: whether the negotiator-side structure changes when player and victim are outgroup-aligned. Used in H2 Bystander only.
- `decision_accept:judged_*` [Acc x N1 ...]: H3 decision-by-judged-status interaction block. Used in H3.
- `iri_*:judged_*` [FS / EC / PT / PD x N1 ...]: H3 empathy-by-judged-status interaction block. Used in H3.
- `participant_engineering` [Eng part.]: Participant faculty contrast. Used in H1/H2/H3.
- `sex_man` [Man]: Sex contrast with woman as the reference. Used in H1/H2/H3.
- `age` [Age]: Participant age in original units. Used in H1/H2/H3.
- `economic_status` [SES]: Economic status in original units. Used in H1/H2/H3.
- `factor(negotiator_slot)` [Slot 2]: Negotiator slot contrast with slot 1 as the reference. Used in H1/H2/H3.

2.2 Interpretation of Interaction Terms

The models herein employ several predefined predictors. It is important to note how interaction terms are interpreted in the context of this behavioral experiment:

1. **Interaction Subsumption:** When an interaction term is statistically significant, it indicates that the effect of one variable depends on the level of the other. Crucially, if the interaction is significant but the constituent main effects are not explicitly significant, their effects are fully subsumed and contextualized by the interaction.
2. **Continuous by Discrete Interactions:** For terms like `iri_total:judged_outgroup`, a negative coefficient implies that the severity of moral judgment (lower score) induced by higher empathy is steeper (magnified) when evaluating an outgroup negotiator relative to the control-labeled baseline. A positive coefficient would mean empathy makes judgments less severe for that outgroup condition.
3. **Discrete by Discrete Interactions:** For terms like `judged_outgroup:decision_accept`, a positive coefficient implies that the change in moral judgment when moving from rejecting a deal to accepting a deal is more positive (less morally condemned) for an outgroup negotiator than for the control-labeled baseline condition.

3 Hypotheses to Test

- **H1 (Empathy Effect):** Empathy dimensions are associated with moral judgment severity after conditioning on judged-negotiator status, counterpart status, decision outcome, and observer-side victim alignment when applicable.
- **H2 (Negotiator-Side Structure):** Moral judgments vary with the joint judged-plus-counterpart negotiator structure, and in the bystander subset that structure may also depend on player-victim alignment.
- **H3 (Empathy x Judged Status):** The empathy effect depends on the judged negotiator’s ingroup, outgroup, or control status after retaining decision context and relational controls.

4 Mathematical Approach and Theoretical Foundations

The analysis employs two complementary estimators for bounded moral judgments. The primary specification is a Two-Limit Tobit model, which is theoretically appropriate for dependent variables that are strictly bounded within a known interval. In this context, moral judgments (y_{isjr}) are observed on a scale from -9 to 9. The Tobit model assumes the existence of a latent, unobserved preference index (y_{isjr}^*) that follows a linear relationship. Option 2 replaces isolated ingroup/outgroup indicators with explicit judgment-level relational variables built from the paired-group structure of each judgment. The analytic hypothesis models decompose that structure into judged-negotiator status, counterpart-negotiator status, observer-side victim alignment when applicable, and hypothesis-specific interaction blocks. H1, H2, and H3 are all estimated separately in the victim and bystander subsets. H1 and H3 use subset-specific formulas so observer-only victim-alignment predictors are retained only where they vary, while H2 uses explicit subset-specific relational-structure blocks:

Victim-subset H2:

$$y_{isj,Victim}^* = \beta_0 + \beta_1 \text{Empathy}_i + \gamma' \mathbf{S}_{isj}^{(V)} + \delta' \mathbf{Z}_i + \epsilon_{isj}$$

Bystander-subset H2:

$$y_{isj,Obs}^* = \beta_0 + \beta_1 \text{Empathy}_i + \gamma' \mathbf{S}_{isj}^{(O)} + \eta V_{is} + \boldsymbol{\theta}'(\mathbf{S}_{isj}^{(O)} \times V_{is}) + \boldsymbol{\delta}' \mathbf{Z}_i + \epsilon_{isj}$$

H3:

$$y_{isjr}^* = \beta_0 + \beta_1 \text{Empathy}_i + \beta_2' G_{isjr} + \beta_3 A_{is} + \beta_4'(G_{isjr} \times A_{is}) + \beta_5'(\text{Empathy}_i \times G_{isjr}) + \beta_6' C_{isjr} + \epsilon_{isjr}, \quad \epsilon_{isjr} \sim N(0, \sigma^2)$$

The actual observed judgment y_{isjr} relates to this latent variable via the censoring transformation:

$$y_{isjr} = \max(-9, \min(9, y_{isjr}^*))$$

This approach prevents the 'ceiling' and 'floor' effects from biasing the linear coefficients, as would occur in standard OLS regression.

5 Analysis of Sample Size Impact

The statistical validity of our inferences depends on the trade-off between False Positives (Type I error, α) and False Negatives (Type II error, β). Given the clustered nature of our data, sample size impact is not merely a count of observations, but a function of cluster correlation.

5.0.1 Step-by-Step Sensitivity Analysis

To determine the impact of our sample size on our ability to detect effects, we follow these steps:

1. Calculate the Design Effect (Deff): As multiple judgments are nested within individuals, we adjust for the Intraclass Correlation (ICC).

$$Deff = 1 + (m - 1) \times ICC$$

where m is the average number of scenarios per participant.

2. Determine the Effective Sample Size (ESS): The ESS represents the number of independent observations that would provide the same statistical power as our clustered sample.

$$ESS = \frac{n_{total}}{Deff}$$

3. Inference Impact: A higher ICC reduces the ESS, thereby increasing the Standard Error of our Tobit coefficients. If the ESS is low, our models become 'conservative', increasing the risk of Type II errors (failing to support a hypothesis). By using clustered robust standard errors for Tobit, we ensure that both sets of p-values acknowledge this reduced information density, protecting the integrity of our Type I error threshold ($\alpha = 0.05$).

Based on the current run, the observed Intraclass Correlation (ICC) is 0.088, with an Effective Sample Size (ESS) of 1819.3.

6 Descriptive Statistics

The empathy profile of the sample is visualized in Figure 1, showing the average scores across the four IRI subscales (Fantasy, Empathic Concern, Perspective Taking, Personal Distress), each scored 0–4 (scale mean per subscale).

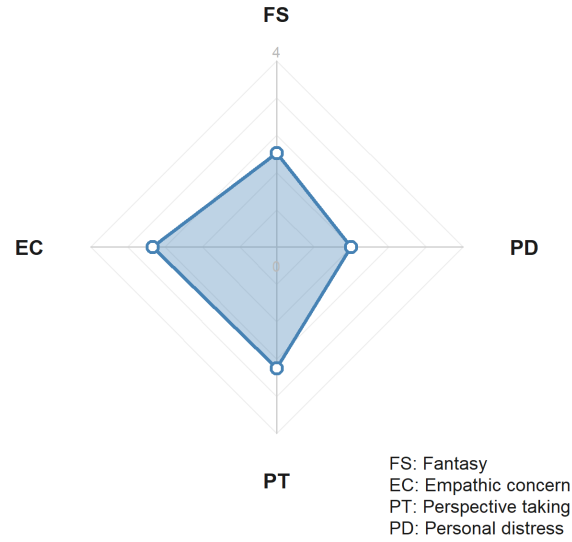


Figure 1: IRI subscale averages (radar plot; scale 0–4 per subscale).

Figure 2 shows severity distributions pooled across the full analytical sample, broken down by N1 group membership (*group_target*: Ingroup, Outgroup, Control). Figures 3 and 4 replicate this breakdown separately within the victim and bystander subsets. Figure 5 adds the bystander-subset view by observer–victim group alignment (*obs_group*), and Figure 6 shows the N2 group panels (*group_other*) for the full sample.

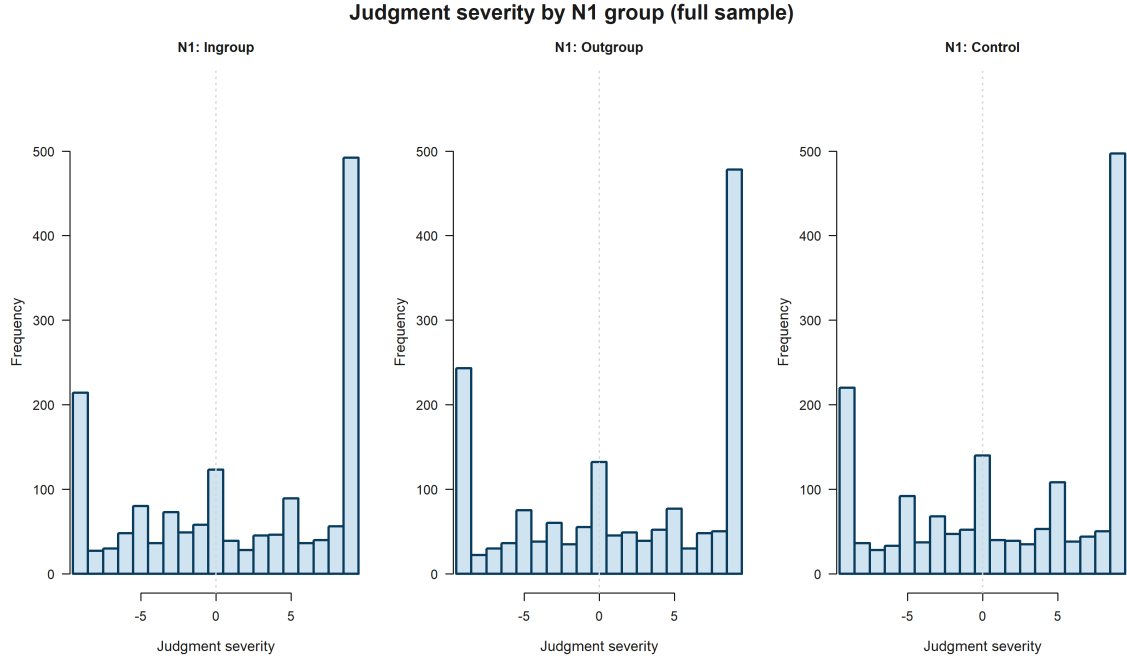


Figure 2: Judgment severity by N1 group (*group_target*: Ingroup, Outgroup, Control) — full analytical sample, scale -9 to 9 .

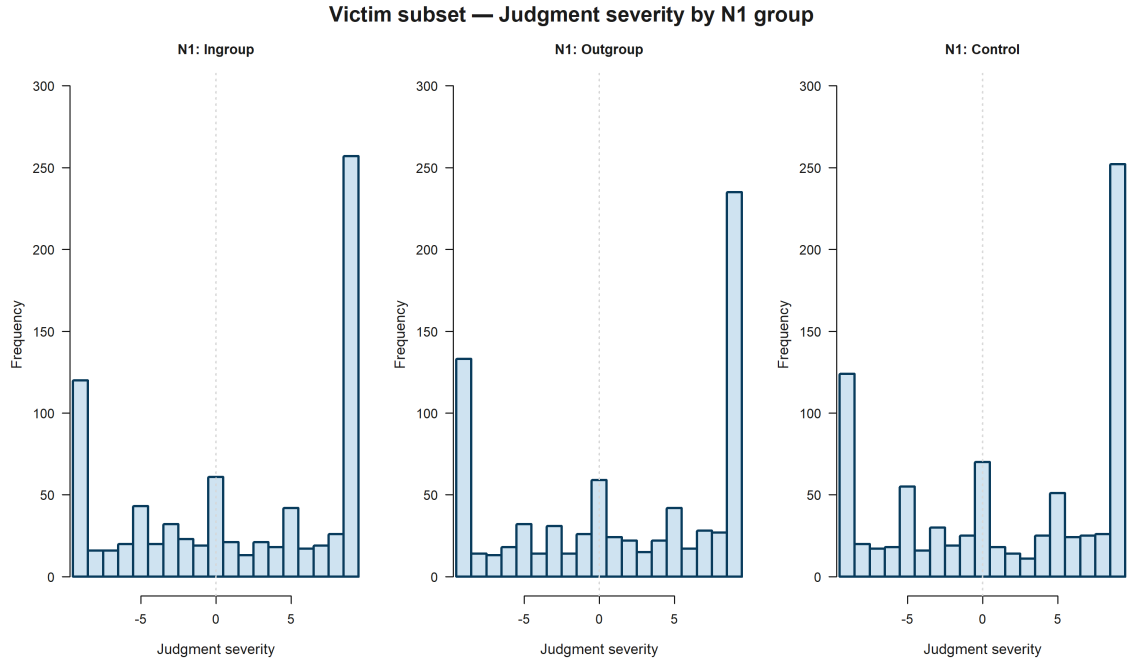


Figure 3: Victim-subset severity by N1 group (*group_target*), scale -9 to 9 .

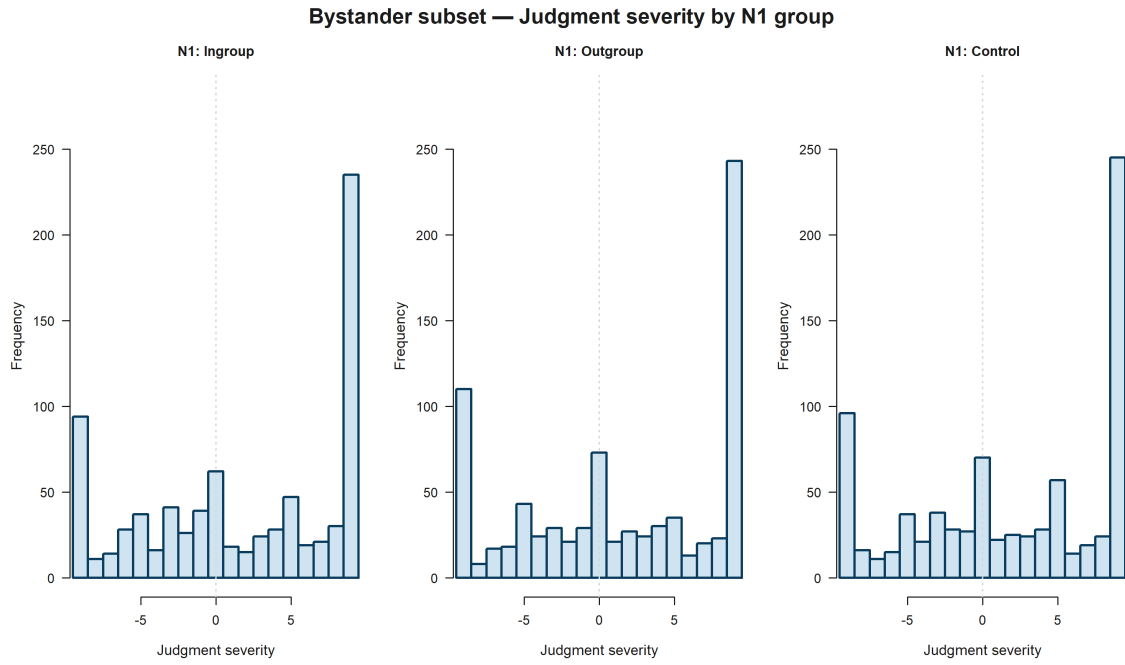


Figure 4: Bystander-subset severity by N1 group (*group_target*), scale -9 to 9 .

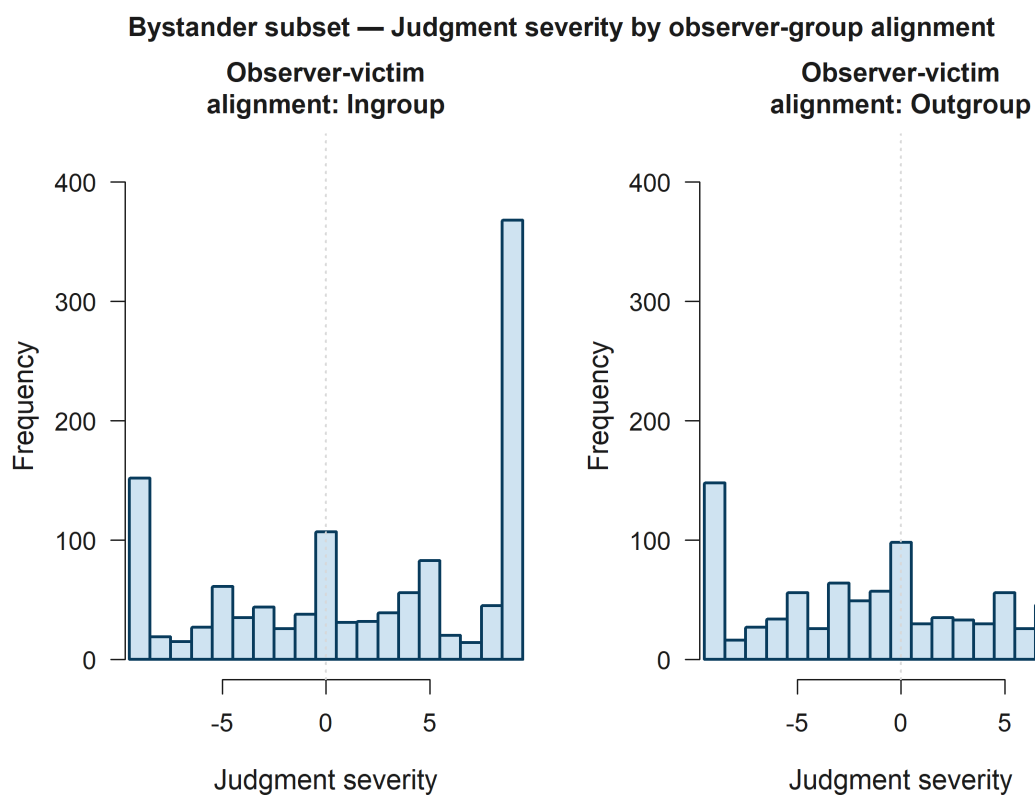


Figure 5: Bystander-subset severity by observer–victim group alignment (*obs_group*: Ingroup, Outgroup), scale -9 to 9 .

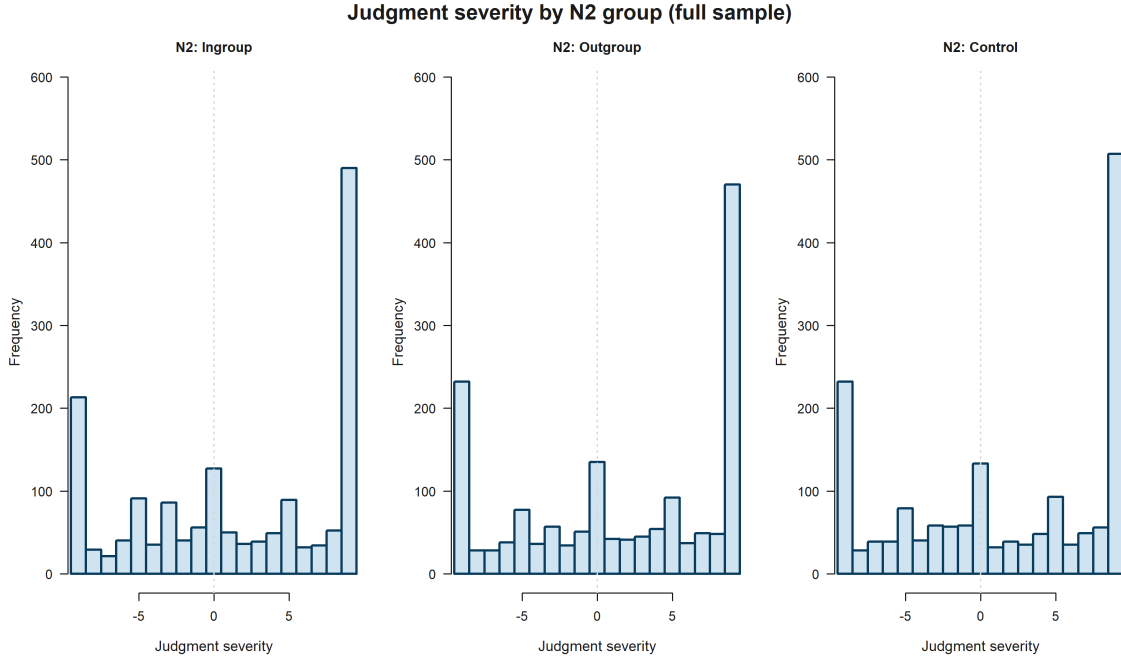


Figure 6: Judgment severity by N2 group (*group_other*: Ingroup, Outgroup, Control) — full analytical sample, scale -9 to 9 .

When a control-labeled negotiator condition exists, the active hypothesis models treat that control level as the reference category. Observer-victim alignment remains binary and therefore keeps ingroup as its reference level.

7 Bi-variate Statistics

The correlation matrix between the psychometric subscales and the mean moral judgment is presented below.

Table 2: Correlation Matrix: IRI Subscales and Moral Judgment.

Fantasy	Empathic Concern	Perspective Taking	Personal Distress	Mean Judgment
1.000	0.476	0.524	0.490	-0.062
0.476	1.000	0.402	0.598	-0.150
0.524	0.402	1.000	0.265	0.008
0.490	0.598	0.265	1.000	-0.003
-0.062	-0.150	0.008	-0.003	1.000

8 Estimator Fit Summary

The following table consolidates the fit-status information for the primary Tobit models used in this report.

Table 3: Estimator fit summary across Tobit specifications.

Model	Approach	Status	Iterations	Observations	Participants	ClusterUnit
H1_B_Bystander_Constructs_Tobit	Tobit	completed	4	2430	243	id
H1_B_Victim_Constructs_Tobit	Tobit	completed	4	2430	243	id
H2_B_Bystander_Constructs_Tobit	Tobit	completed	3	2430	243	id
H2_B_Victim_Constructs_Tobit	Tobit	completed	3	2430	243	id
H3_B_Bystander_Constructs_Tobit	Tobit	completed	4	2430	243	id
H3_B_Victim_Constructs_Tobit	Tobit	completed	4	2430	243	id

8.1 Hypothesis Significance Summary

The following concise tables list only hypothesis-relevant predictors that reached at least $p < 0.10$ in the available Tobit models, split by victim and bystander subsets, using conventional symbols to indicate strength of evidence (+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Dynamic figures are generated only for predictors that appear in these tables with at least one significance symbol.

Table 4: Hypothesis-level significance summary for the victim subset across Tobit models.

Hypothesis	Tobit support
H1	PT*
H2	None
H3	PT x N1-In*; PT x N1-Out+

Victim subset

Note. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Table 5: Hypothesis-level significance summary for the bystander subset across Tobit models.

Hypothesis	Tobit support
H1	None
H2	N1 Out, N2 In+
H3	None

Bystander subset

Note. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

8.2 Significance-Driven Figures

Only hypothesis-relevant predictors that reach $p < 0.10$ or better are visualized automatically. These dynamic figures use the saved Tobit outputs, and participant id remains only an inference-level clustering unit rather than a substantive explanatory variable.

Empathy Effect Empathy: Perspective taking is statistically significant in the Tobit model (*, $p = 0.010$). Figure 7 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted judgment.

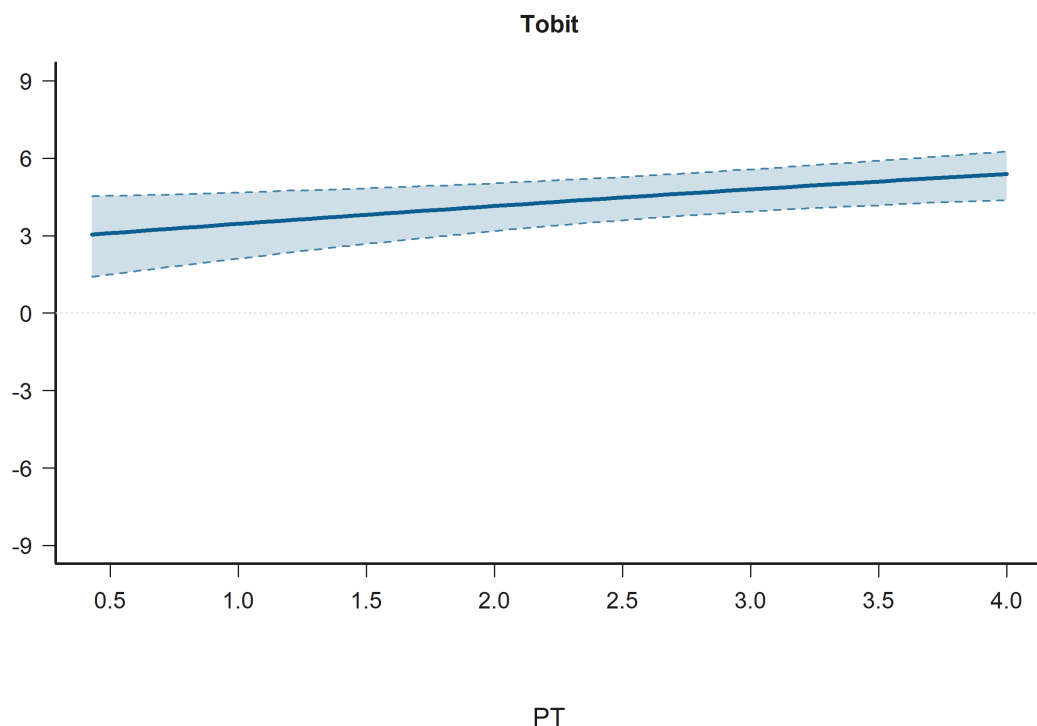


Figure 7: Effect Plot for Empathy: Perspective taking in Empathy Effect. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Negotiator-Side Structure Negotiator-side structure: N1 outgroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) is statistically significant in the Tobit model (+, $p = 0.054$). Figure 8 shows that predicted judgment is lower for N1 Out, N2 In than for Ref: N1 Ctl, N2 Ctl.



Figure 8: Grouped Prediction Plot for Negotiator-side structure: N1 outgroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) in Negotiator-Side Structure. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy x Judged Status Empathy: Perspective taking x N1 ingroup is statistically significant in the Tobit model (*, $p = 0.040$). Figure 9 shows that the predicted relationship rises most sharply for Empathy: Perspective taking when the condition is N1 In.

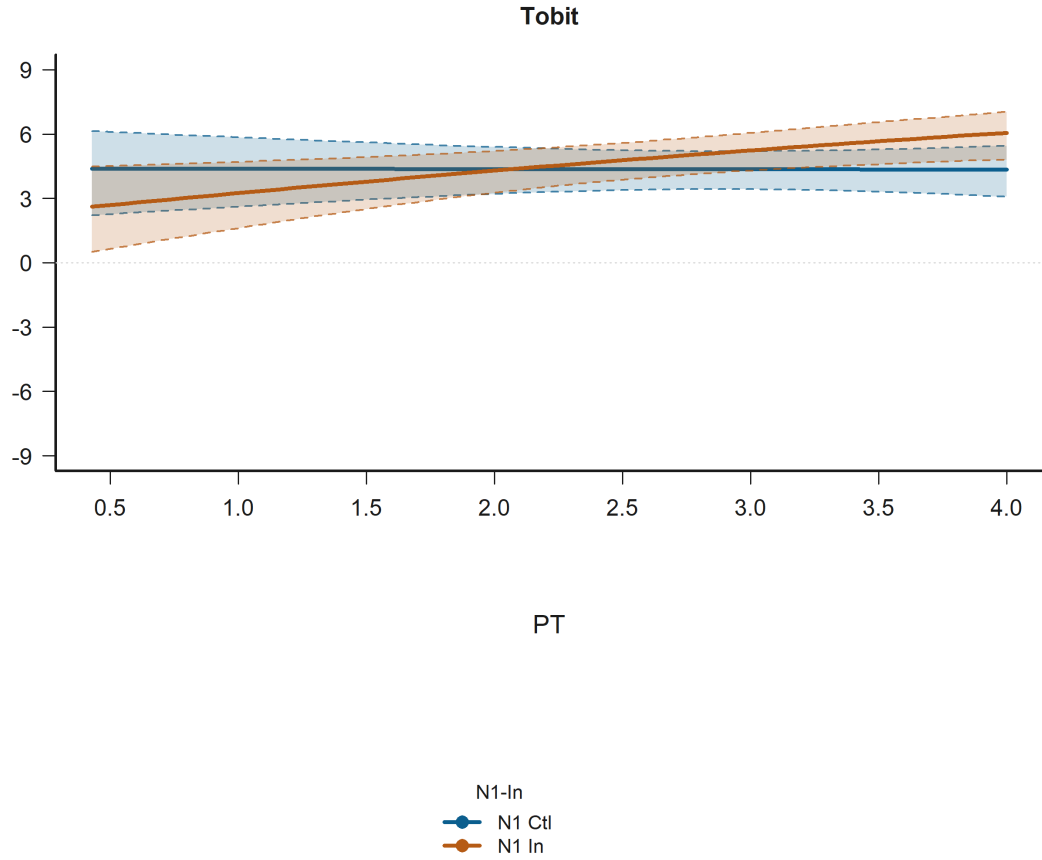


Figure 9: Interaction Plot for Empathy: Perspective taking x N1 ingroup in Empathy x Judged Status. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Perspective taking x N1 outgroup is statistically significant in the Tobit model (+, $p = 0.050$). Figure 10 shows that the predicted relationship rises most sharply for Empathy: Perspective taking when the condition is N1 Out.

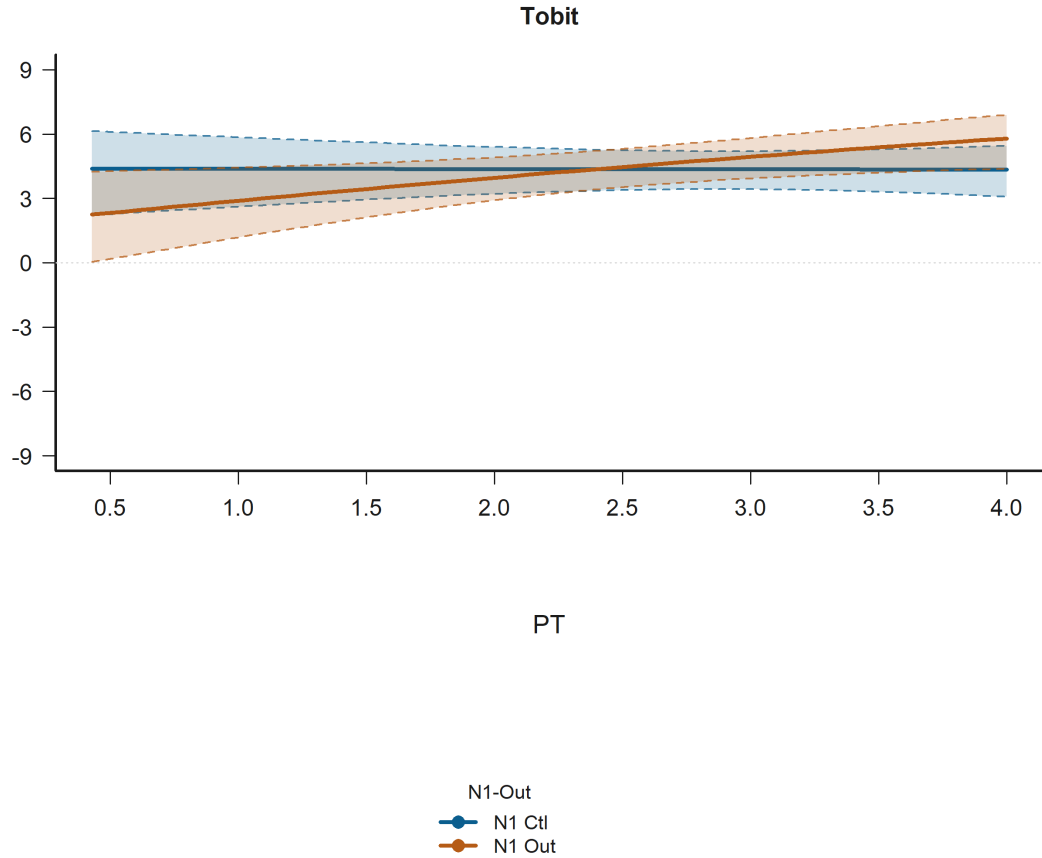


Figure 10: Interaction Plot for Empathy: Perspective taking x N1 outgroup in Empathy x Judged Status. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

8.3 All Significant Predictors ($p < 0.10$)

The following figures extend beyond the hypothesis-target terms and visualize every predictor that reaches $p < 0.10$ in the available H1-H3 Tobit models. This includes significant controls such as age when they clear the threshold.

judgments_bystander Harm Accepted is statistically significant in:

- Empathy Effect Model B (Tobit)
- Empathy x Judged Status Model B (Tobit)

Figure 11 shows that predicted judgment is lower for Acc than for Rej.



Figure 11: Grouped Prediction Plot for Harm Accepted in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Age is statistically significant in:

- Negotiator-Side Structure Model B (Tobit)
- Empathy x Judged Status Model B (Tobit)
- Empathy Effect Model B (Tobit)

Figure 12 shows that across the observed range, higher Age corresponds to higher predicted judgment.

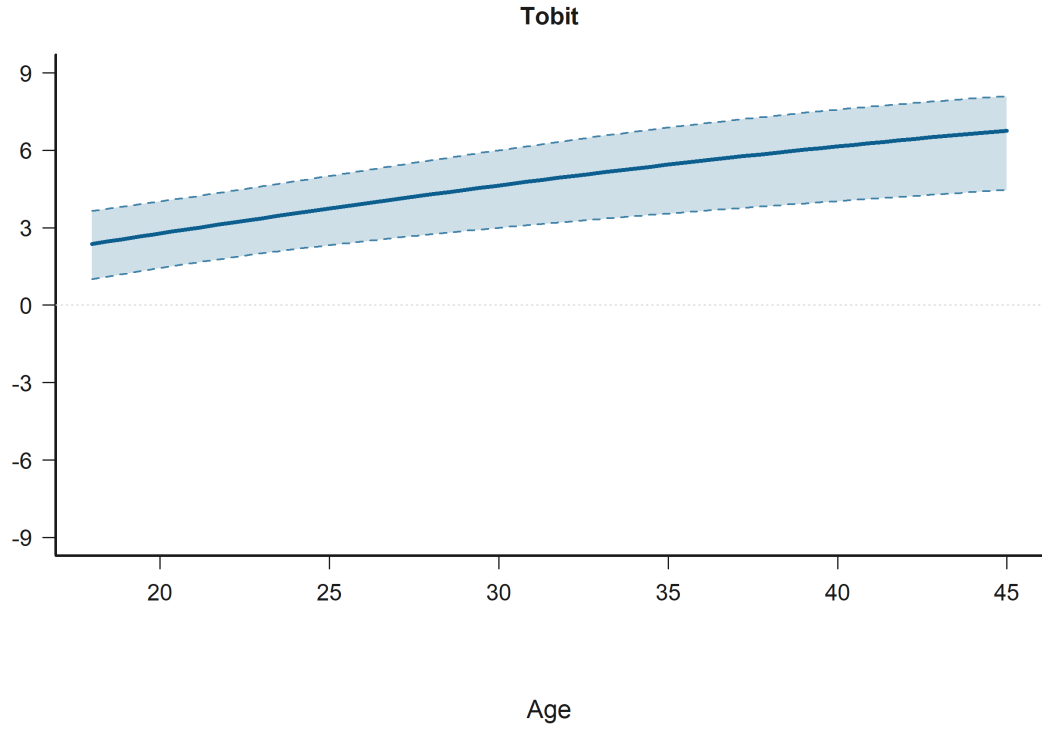


Figure 12: Effect Plot for Age in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

N1 outgroup is statistically significant in:

- Empathy x Judged Status Model B (Tobit)

Figure 13 shows that predicted judgment is lower for N1 Out than for N1 Ctl.



Figure 13: Grouped Prediction Plot for N1 outgroup in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

N1 ingroup is statistically significant in:

- Empathy x Judged Status Model B (Tobit)

Figure 14 shows that predicted judgment is lower for N1 In than for N1 Ctl.



Figure 14: Grouped Prediction Plot for N1 ingroup in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Negotiator-side structure: N1 outgroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) is statistically significant in:

- Negotiator-Side Structure Model B (Tobit)

Figure 15 shows that predicted judgment is lower for N1 Out, N2 In than for Ref: N1 Ctl, N2 Ctl.



Figure 15: Grouped Prediction Plot for Negotiator-side structure: N1 outgroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Engineering participant is statistically significant in:

- Empathy Effect Model B (Tobit)
- Empathy x Judged Status Model B (Tobit)
- Negotiator-Side Structure Model B (Tobit)

Figure 16 shows that predicted judgment is higher for Eng part. than for Hum part..



Figure 16: Grouped Prediction Plot for Engineering participant in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Perspective taking is statistically significant in:

- Empathy x Judged Status Model B (Tobit)

Figure 17 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted judgment.

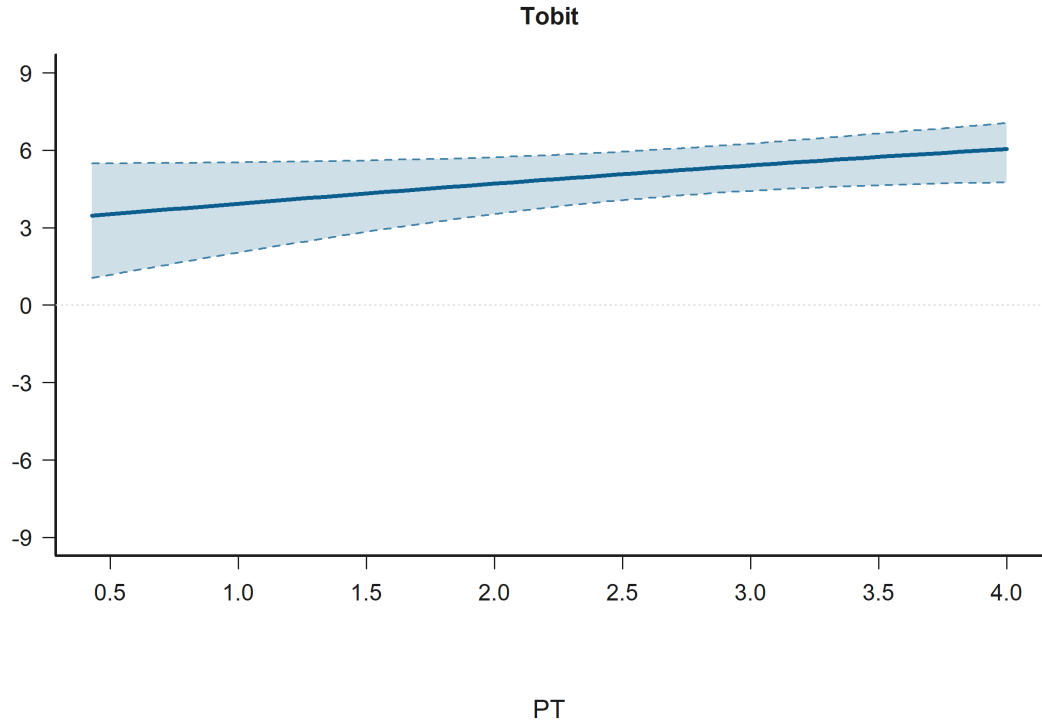


Figure 17: Effect Plot for Empathy: Perspective taking in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

judgments_victim Harm Accepted is statistically significant in:

- Empathy Effect Model B (Tobit)
- Empathy x Judged Status Model B (Tobit)

Figure 18 shows that predicted judgment is lower for Acc than for Rej.



Figure 18: Grouped Prediction Plot for Harm Accepted in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Perspective taking is statistically significant in:

- Negotiator-Side Structure Model B (Tobit)
- Empathy Effect Model B (Tobit)

Figure 19 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted judgment.

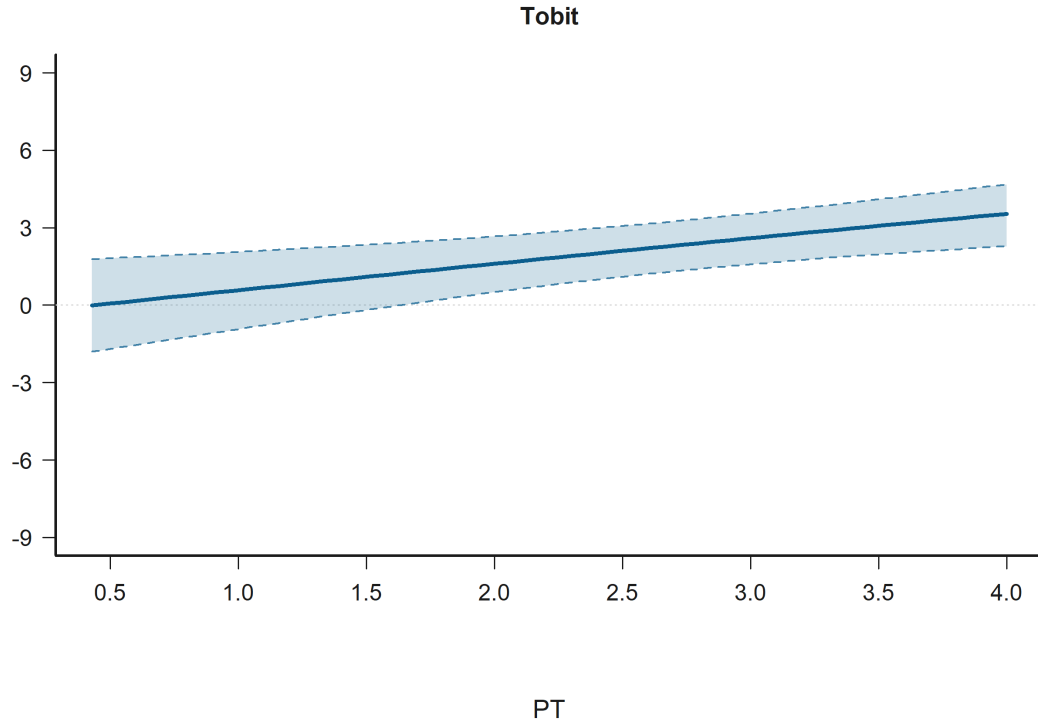


Figure 19: Effect Plot for Empathy: Perspective taking in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Perspective taking x N1 ingroup is statistically significant in:

- Empathy x Judged Status Model B (Tobit)

Figure 20 shows that the predicted relationship rises most sharply for Empathy: Perspective taking when the condition is N1 In.

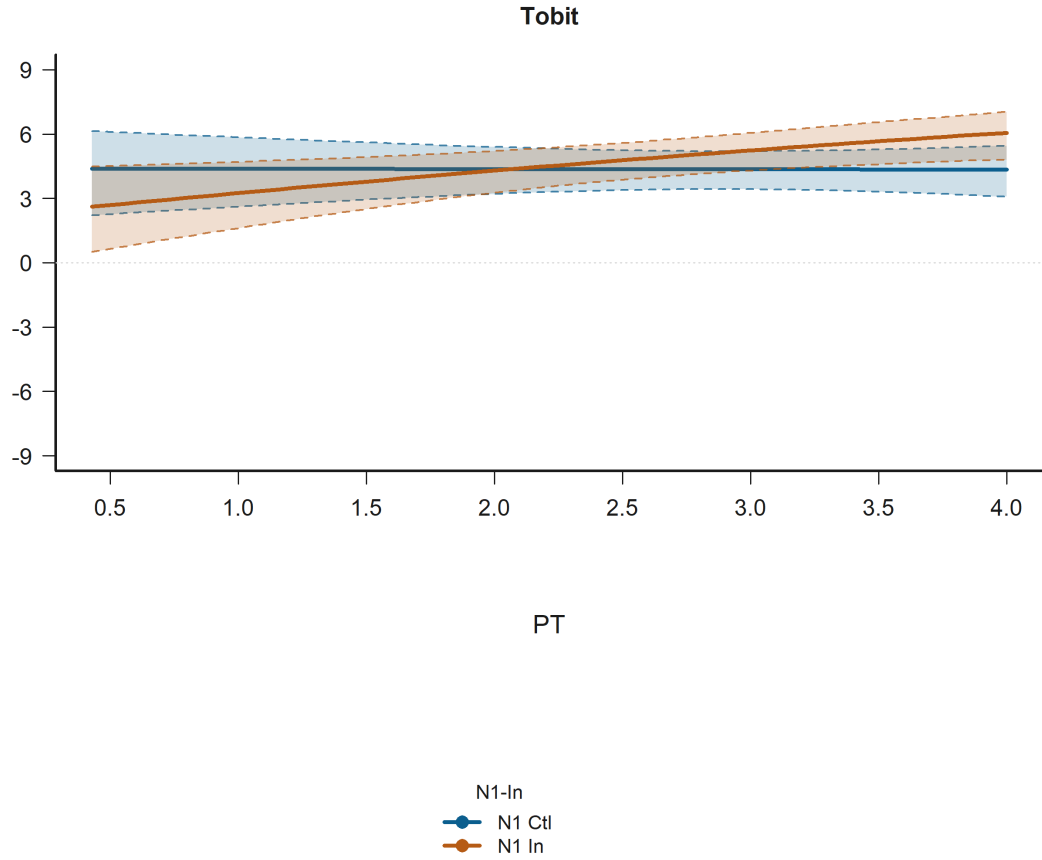


Figure 20: Interaction Plot for Empathy: Perspective taking x N1 ingroup in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Perspective taking x N1 outgroup is statistically significant in:

- Empathy x Judged Status Model B (Tobit)

Figure 21 shows that the predicted relationship rises most sharply for Empathy: Perspective taking when the condition is N1 Out.

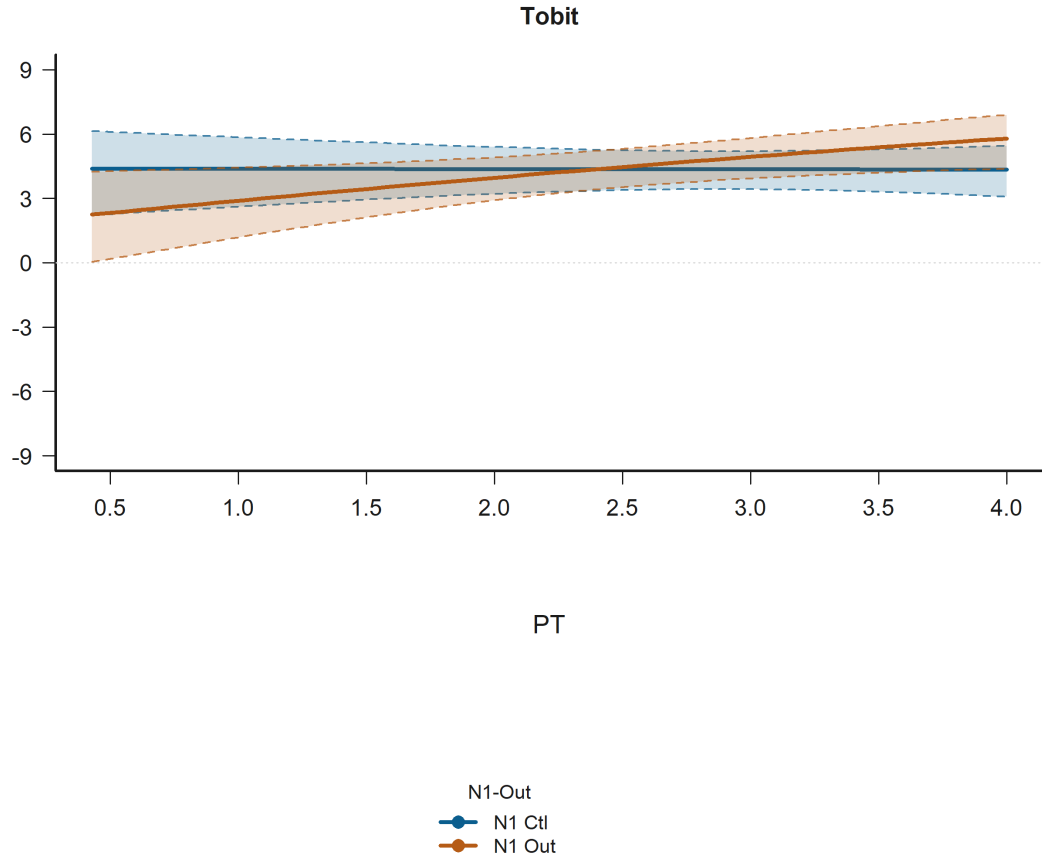


Figure 21: Interaction Plot for Empathy: Perspective taking x N1 outgroup in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Engineering participant is statistically significant in:

- Negotiator-Side Structure Model B (Tobit)

Figure 22 shows that predicted judgment is higher for Eng part. than for Hum part..

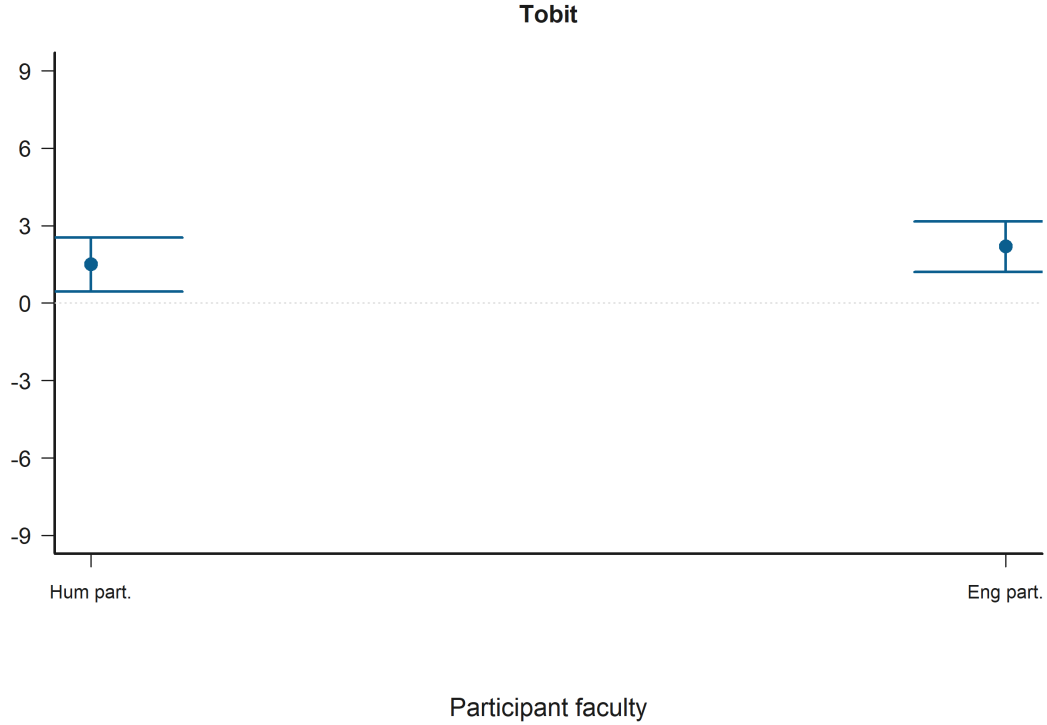


Figure 22: Grouped Prediction Plot for Engineering participant in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

8.4 Integrated Hypothesis Conclusions

The following summary restates each original hypothesis and indicates whether the available Tobit estimates support it in the current data.

- H1. Original hypothesis: Empathy dimensions are associated with moral judgment severity after conditioning on judged-negotiator status, counterpart status, decision outcome, and observer-side victim alignment when applicable. Victim subset: Tobit conclusion: the available models point in the opposite direction of the hypothesis. Model B points in the opposite direction through Empathy: Perspective taking with a positive association ($p = 0.010$). Additional statistically significant signals include Harm Accepted with a negative association ($p < 0.001$) and Empathy: Perspective taking with a positive association ($p = 0.010$). Bystander subset: Tobit conclusion: the available models do not support the hypothesis. Model B does not support the hypothesis; none of four empathy dimensions are statistically significant, and the closest signal is Empathy: Perspective taking with a positive association ($p = 0.163$). Additional statistically significant signals include Harm Accepted with a negative association ($p < 0.001$) and Age with a positive association ($p = 0.005$).

- H2. Original hypothesis: Moral judgments vary with the joint judged-plus-counterpart negotiator structure, and in the bystander subset that structure may also depend on player-victim alignment. Victim subset: Tobit conclusion: the available models do not support the hypothesis. Model B does not support the hypothesis; none of negotiator-side structure contrasts are statistically significant, and the closest signal is Negotiator-side structure: N1 control label hidden, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) with a negative association ($p = 0.227$). Additional statistically significant signals include Empathy: Perspective taking with a positive association ($p = 0.003$) and Engineering participant with a positive association ($p = 0.065$). Bystander subset: Tobit conclusion: the available models support the hypothesis. Model B supports the hypothesis through Negotiator-side structure: N1 outgroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) with a negative association ($p = 0.054$). Additional statistically significant signals include Age with a positive association ($p < 0.001$) and Negotiator-side structure: N1 outgroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) with a negative association ($p = 0.054$).
- H3. Original hypothesis: The empathy effect depends on the judged negotiator's ingroup, outgroup, or control status after retaining decision context and relational controls. Victim subset: Tobit conclusion: the available models support the hypothesis. Model B supports the hypothesis through Empathy: Perspective taking x N1 ingroup with a positive association ($p = 0.040$) and Empathy: Perspective taking x N1 outgroup with a positive association ($p = 0.050$). Additional statistically significant signals include Harm Accepted with a negative association ($p < 0.001$) and Empathy: Perspective taking x N1 ingroup with a positive association ($p = 0.040$). Bystander subset: Tobit conclusion: the available models do not support the hypothesis. Model B does not support the hypothesis; none of empathy-by-judged-status interactions are statistically significant, and the closest signal is Empathy: Perspective taking x N1 ingroup with a negative association ($p = 0.111$). Additional statistically significant signals include Harm Accepted with a negative association ($p < 0.001$) and Age with a positive association ($p = 0.004$).

9 Hypothesis Validation and Results

Detailed coefficient tables for each Tobit model, including predictor estimates, standard errors, p-values with conventional significance symbols, and estimator diagnostics for the victim and bystander subsets, are provided below.

9.1 H1: Empathy Effect

This section evaluates H1 separately in the victim and bystander subsets using the four IRI subscales as the active empathy specification: fantasy, empathic concern, perspective taking, and personal distress. The formula retains judged-negotiator status, counterpart status, decision outcome, and participant controls for sex, age, and economic status, while the bystander subset additionally includes the observer-side victim-alignment predictor because it is meaningful only in that subset.

9.1.1 Active Empathy-Construct Specification

Model B subset-specific formulas.

Victim: judgement
 $\text{textasciitilde iri_fs} + \text{iri_ec} + \text{iri_pt} + \text{iri_pd} + \text{judged_ingroup} + \text{judged_outgroup} +$

counterpart_ingroup + counterpart_outgroup + decision_accept + participant_engineering
+ sex_man + age + economic_status + factor(negotiator_slot)

Bystander: judgement

textasciitilde iri_fs + iri_ec + iri_pt + iri_pd + judged_ingroup + judged_outgroup + counterpart_ingroup + counterpart_outgroup + decision_accept + observer_victim_outgroup + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Victim subset

Table 6: H1 active model: empathy-construct Tobit coefficients (Tobit estimator; Victim subset: 2,430 observations from 243 participants).

Predictor	Estimate	Std. Error	p-value
Base	3.525	3.001	0.240
FS	0.358	0.485	0.461
EC	-0.283	0.664	0.670
PT	1.317	0.514	0.010*
PD	-0.153	0.568	0.788
N1-In	0.303	0.537	0.573
N1-Out	-0.166	0.552	0.764
N2-In	-0.148	0.578	0.798
N2-Out	-0.168	0.552	0.761
Acc	-9.647	0.796	<0.001***
Eng part.	0.879	0.621	0.157
Man	-0.317	0.663	0.633
Age	-0.000	0.127	0.997
SES	0.283	0.294	0.336
Slot 2	-0.151	0.447	0.736
Log(s)	2.344	0.054	<0.001***

Tobit estimator

Note. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The **Baseline judgment** represents the starting moral score when all continuous variables are at their average and all group variables are at their reference levels (for the active negotiator-status terms, the control-labeled condition). In this model, that baseline is estimated at 3.525 (p=0.240). The variable **Empathy: Fantasy scale** (the tendency to imaginatively get involved in fictitious stories (Fantasy subscale)) has an estimated effect of 0.358. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.36 points on the -9 to 9 scale. The variable **Empathy: Empathic concern** (the tendency to feel sympathy and compassion for others (Empathic Concern subscale)) has an estimated effect of -0.283. The results show indicates

little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.28 points on the -9 to 9 scale. The variable ****Empathy: Perspective taking**** (the tendency to look at things from another's point of view (Perspective Taking subscale)) has an estimated effect of 1.317. The results show indicates clear evidence of an effect, showing that this variable leads to an increase (higher scores) in the moral judgment. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 1.32 points on the -9 to 9 scale. The variable ****Empathy: Personal distress**** (the tendency to feel distress in uncomfortable social situations (Personal Distress subscale)) has an estimated effect of -0.153. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.15 points on the -9 to 9 scale. The variable ****N1 ingroup**** (whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of 0.303. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.30 points on the -9 to 9 scale. The variable ****N1 outgroup**** (whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -0.166. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.17 points on the -9 to 9 scale. The variable ****N2 ingroup**** (whether N2 was ingroup rather than in the control-labeled baseline) has an estimated effect of -0.148. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.15 points on the -9 to 9 scale. The variable ****N2 outgroup**** (whether N2 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -0.168. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.17 points on the -9 to 9 scale. The variable ****Harm Accepted**** (whether N1 accepted the harmful deal) has an estimated effect of -9.647. The results show indicates very strong evidence of an effect, showing that this variable leads to a decrease (lower, more severe scores) in the moral judgment. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -9.65 points on the -9 to 9 scale. The variable ****Engineering participant**** (whether the participant is from an Engineering faculty) has an estimated effect of 0.879. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.88 points on the -9 to 9 scale. The variable ****Man participant**** (being a man (relative to being a woman)) has an estimated effect of -0.317. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.32 points on the -9 to 9 scale. The variable ****Age**** (the participant's age) has an estimated effect of -0.000. The results show indicates little to no evidence of an effect, meaning

the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.00 points on the -9 to 9 scale. The variable ****Economic status**** (the socioeconomic contextual stratum of the participant's background) has an estimated effect of 0.283. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.28 points on the -9 to 9 scale. The ****Judgment variation (log-scale)**** is 2.344, which indicates the level of dispersion or 'noise' in the latent judgments. This reflects the inherent variety of moral opinions across the sample.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.959$, $p = 1.041e-25$). The primary Tobit model still uses participant-clustered standard errors by id,.

Bystander subset

Table 7: H1 active model: empathy-construct Tobit coefficients (Tobit estimator; Bystander subset: 2,430 observations from 243 participants).

Predictor	Estimate	Std. Error	p-value
Base	-0.311	2.720	0.909
FS	0.408	0.395	0.301
EC	-0.834	0.603	0.167
PT	0.696	0.499	0.163
PD	0.074	0.498	0.882
N1-In	-0.380	0.494	0.442
N1-Out	-0.366	0.505	0.468
N2-In	0.084	0.452	0.852
N2-Out	-0.168	0.477	0.724
Acc	-8.681	0.696	<0.001***
V-Out	-0.129	0.434	0.766
Eng part.	1.025	0.543	0.059+
Man	0.337	0.599	0.574
Age	0.289	0.103	0.005**
SES	0.230	0.274	0.402
Slot 2	0.603	0.428	0.159
Log(s)	2.223	0.054	<0.001***

Tobit estimator

Note. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The ****Baseline judgment**** represents the starting moral score when all continuous variables are at their average and all group variables are at their reference levels (for the active negotiator-status terms, the control-labeled condition). In this model, that baseline is estimated at -0.311 ($p=0.909$). The variable ****Empathy: Fantasy scale**** (the tendency to imaginatively get involved in fictitious stories (Fantasy subscale)) has an estimated effect of 0.408. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.41 points on the -9 to 9 scale. The variable ****Empathy: Empathic concern**** (the tendency to feel sympathy and compassion for others (Empathic Concern subscale)) has an estimated effect of -0.834. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.83 points on the -9 to 9 scale. The variable ****Empathy: Perspective taking**** (the tendency to look at things from another's point of view (Perspective Taking subscale)) has an estimated effect of 0.696. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.70 points on the -9 to 9 scale. The variable ****Empathy: Personal distress**** (the tendency to feel distress in uncomfortable social situations (Personal Distress subscale)) has an estimated effect of 0.074. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.07 points on the -9 to 9 scale. The variable ****N1 ingroup**** (whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of -0.380. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.38 points on the -9 to 9 scale. The variable ****N1 outgroup**** (whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -0.366. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.37 points on the -9 to 9 scale. The variable ****N2 ingroup**** (whether N2 was ingroup rather than in the control-labeled baseline) has an estimated effect of 0.084. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.08 points on the -9 to 9 scale. The variable ****N2 outgroup**** (whether N2 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -0.168. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.17 points on the -9 to 9 scale. The variable ****Harm Accepted**** (whether N1 accepted the harmful deal) has an estimated effect of -8.681. The results show indicates very strong evidence of an effect, showing that this variable leads to a decrease (lower, more severe scores) in the moral judgment. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -8.68 points on the -9 to 9 scale. The variable ****Observer-side victim outgroup**** (whether the victim belonged to a different

faculty than the observer (Outgroup)) has an estimated effect of -0.129. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.13 points on the -9 to 9 scale. The variable ****Engineering participant**** (whether the participant is from an Engineering faculty) has an estimated effect of 1.025. The results show indicates suggestive but inconclusive evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 1.03 points on the -9 to 9 scale. The variable ****Man participant**** (being a man (relative to being a woman)) has an estimated effect of 0.337. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.34 points on the -9 to 9 scale. The variable ****Age**** (the participant's age) has an estimated effect of 0.289. The results show indicates strong evidence of an effect, showing that this variable leads to an increase (higher scores) in the moral judgment. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.29 points on the -9 to 9 scale. The variable ****Economic status**** (the socioeconomic contextual stratum of the participant's background) has an estimated effect of 0.230. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.23 points on the -9 to 9 scale. The ****Judgment variation (log-scale)**** is 2.223, which indicates the level of dispersion or 'noise' in the latent judgments. This reflects the inherent variety of moral opinions across the sample.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.968$, $p = 6.754e-23$). The primary Tobit model still uses participant-clustered standard errors by id,.

9.2 H2: Negotiator-Side Relational Structure

This section estimates H2 separately in the victim and bystander subsets. The dependent variable remains negotiator-specific judgment severity, so each participant contributes two judgment rows per scenario, one for each negotiator. In the victim subset, H2 uses the joint judged-plus-counterpart negotiator structure relative to the victim-player, with the control-labeled N1 plus control-labeled N2 condition as the reference structure. In the bystander subset, the same negotiator-side structure is defined relative to the observing player's faculty, and the model additionally includes the player-victim outgroup term and its interaction with the negotiator-side structure dummies. Victim-subset equation: $y_{isj,Victim}^* = \beta_0 + \beta_1 \text{Empathy}_i + \gamma' \mathbf{S}_{isj}^{(V)} + \delta' \mathbf{Z}_i + \epsilon_{isj}$, where $\mathbf{S}_{isj}^{(V)}$ indexes the judged-plus-counterpart structure dummies and $\mathbf{Z}_i$ collects participant controls. Bystander-subset equation: $y_{isj,Obs}^* = \beta_0 + \beta_1 \text{Empathy}_i + \gamma' \mathbf{S}_{isj}^{(O)} + \eta V_{is} + \theta' (\mathbf{S}_{isj}^{(O)} \times V_{is}) + \delta' \mathbf{Z}_i + \epsilon_{isj}$, where V_{is} is the player-victim outgroup indicator.

9.2.1 Active Empathy-Construct Specification

Model B subset-specific formulas.

Victim: judgement

textasciitilde iri_fs + iri_ec + iri_pt + iri_pd + h2_negstruct_j_in_c_in + h2_negstruct_j_out_c_in
+ h2_negstruct_j_cont_c_in + h2_negstruct_j_in_c_out + h2_negstruct_j_out_c_out
+ h2_negstruct_j_cont_c_out + h2_negstruct_j_in_c_cont + h2_negstruct_j_out_c_cont
+ participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Bystander: judgement

textasciitilde iri_fs + iri_ec + iri_pt + iri_pd + h2_negstruct_j_in_c_in + h2_negstruct_j_out_c_in
+ h2_negstruct_j_cont_c_in + h2_negstruct_j_in_c_out + h2_negstruct_j_out_c_out
+ h2_negstruct_j_cont_c_out + h2_negstruct_j_in_c_cont + h2_negstruct_j_out_c_cont
+ player_victim_outgroup + player_victim_outgroup:h2_negstruct_j_in_c_in + player_victim_outgroup:h2_neg
+ player_victim_outgroup:h2_negstruct_j_cont_c_in + player_victim_outgroup:h2_negstruct_j_in_c_out
+ player_victim_outgroup:h2_negstruct_j_out_c_out + player_victim_outgroup:h2_negstruct_j_cont_c_out
+ player_victim_outgroup:h2_negstruct_j_in_c_cont + player_victim_outgroup:h2_negstruct_j_out_c_cont
+ participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Victim subset

Table 8: H2 active model: empathy-construct Tobit coefficients (Tobit estimator; Victim subset: 2,430 observations from 243 participants).

Predictor	Estimate	Std. Error	p-value
Base	-1.475	3.307	0.656
FS	0.721	0.513	0.159
EC	-0.915	0.756	0.226
PT	1.843	0.612	0.003**
PD	-0.040	0.587	0.946
N1 In, N2 In	0.132	1.063	0.901
N1 Out, N2 In	-0.325	1.057	0.758
N1 Ctl, N2 In	-1.284	1.062	0.227
N1 In, N2 Out	0.035	0.999	0.972
N1 Out, N2 Out	-0.826	1.088	0.448
N1 Ctl, N2 Out	-0.707	1.120	0.528
N1 In, N2 Ctl	-0.749	1.005	0.456
N1 Out, N2 Ctl	-0.800	1.096	0.465
Eng part.	1.278	0.694	0.065+
Man	-0.406	0.769	0.598
Age	-0.027	0.145	0.852
SES	0.466	0.333	0.162
Slot 2	-0.017	0.459	0.970
Log(s)	2.450	0.057	<0.001***

Tobit estimator

Note. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The ****Baseline judgment**** represents the starting moral score when all continuous variables are at their average and all group variables are at their reference levels (for the active negotiator-status terms, the control-labeled condition). In this model, that baseline is estimated at -1.475 ($p=0.656$). The variable ****Empathy: Fantasy scale**** (the tendency to imaginatively get involved in fictitious stories (Fantasy subscale)) has an estimated effect of 0.721. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.72 points on the -9 to 9 scale. The variable ****Empathy: Empathic concern**** (the tendency to feel sympathy and compassion for others (Empathic Concern subscale)) has an estimated effect of -0.915. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.91 points on the -9 to 9 scale. The variable ****Empathy: Perspective taking**** (the tendency to look at things from another's point of view (Perspective Taking subscale)) has an estimated effect of 1.843. The results show indicates strong evidence of an effect, showing that this variable leads to an increase (higher scores) in the moral judgment. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 1.84 points on the -9 to 9 scale. The variable ****Empathy: Personal distress**** (the tendency to feel distress in uncomfortable social situations (Personal Distress subscale)) has an estimated effect of -0.040. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.04 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 ingroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 ingroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of 0.132. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.13 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 outgroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 outgroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of -0.325. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.33 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 control label hidden, N2 ingroup (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 control label hidden, N2 ingroup (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of -1.284. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.28 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 ingroup, N2 outgroup (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 ingroup, N2 outgroup (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of 0.035. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.03 points

on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 outgroup, N2 outgroup (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 outgroup, N2 outgroup (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of -0.826. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.83 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 control label hidden, N2 outgroup (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 control label hidden, N2 outgroup (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of -0.707. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.71 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 ingroup, N2 control label hidden (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 ingroup, N2 control label hidden (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of -0.749. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.75 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 outgroup, N2 control label hidden (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 outgroup, N2 control label hidden (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of -0.800. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.80 points on the -9 to 9 scale. The variable ****Engineering participant**** (whether the participant is from an Engineering faculty) has an estimated effect of 1.278. The results show indicates suggestive but inconclusive evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 1.28 points on the -9 to 9 scale. The variable ****Man participant**** (being a man (relative to being a woman)) has an estimated effect of -0.406. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.41 points on the -9 to 9 scale. The variable ****Age**** (the participant's age) has an estimated effect of -0.027. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.03 points on the -9 to 9 scale. The variable ****Economic status**** (the socioeconomic contextual stratum of the participant's background) has an estimated effect of 0.466. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.47 points on the -9 to 9 scale. The ****Judgment variation (log-scale)**** is 2.450, which indicates the level of dispersion or 'noise' in the latent judgments. This reflects the inherent variety of moral opinions across the sample.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.914$, $p = 5.152e-35$). The primary Tobit model still uses participant-clustered standard errors by id,.

Bystander subset

Table 9: H2 active model: empathy-construct Tobit coefficients (Tobit estimator; Bystander subset: 2,430 observations from 243 participants).

Predictor	Estimate	Std. Error	p-value
Base	-5.190	2.968	0.080+
FS	0.222	0.444	0.617
EC	-0.861	0.650	0.185
PT	0.704	0.521	0.177
PD	0.115	0.564	0.838
N1 In, N2 In	0.129	1.481	0.930
N1 Out, N2 In	-2.666	1.381	0.054+
N1 Ctl, N2 In	0.257	1.406	0.855
N1 In, N2 Out	-1.548	1.441	0.283
N1 Out, N2 Out	-1.194	1.564	0.445
N1 Ctl, N2 Out	-1.042	1.440	0.469
N1 In, N2 Ctl	-0.837	1.415	0.554
N1 Out, N2 Ctl	-0.568	1.518	0.708
V-Out	-0.414	1.499	0.782
Eng part.	0.942	0.572	0.099+
Man	0.739	0.641	0.249
Age	0.349	0.101	<0.001***
SES	0.338	0.293	0.249
Slot 2	0.729	0.446	0.102
N1 In, N2 In x V-Out	-1.441	2.064	0.485
N1 Out, N2 In x V-Out	1.869	1.954	0.339
N1 Ctl, N2 In x V-Out	-0.325	1.796	0.856
N1 In, N2 Out x V-Out	1.253	2.059	0.543
N1 Out, N2 Out x V-Out	2.518	2.269	0.267
N1 Ctl, N2 Out x V-Out	-1.777	1.922	0.355
N1 In, N2 Ctl x V-Out	-1.342	1.978	0.497
N1 Out, N2 Ctl x V-Out	-1.091	2.018	0.589
Log(s)	2.328	0.057	<0.001***

Tobit estimator

Note. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The ****Baseline judgment**** represents the starting moral score when all continuous variables are at their average and all group variables are at their reference levels (for the active negotiator-status terms, the control-labeled condition). In this model, that baseline is estimated at -5.190 ($p=0.080$). The variable ****Empathy: Fantasy scale**** (the tendency to imaginatively get involved in fictitious stories (Fantasy subscale)) has an estimated effect of 0.222. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.22 points on the -9 to 9 scale. The variable ****Empathy: Empathic concern**** (the tendency to feel sympathy and compassion for others (Empathic Concern subscale)) has an estimated effect of -0.861. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.86 points on the -9 to 9 scale. The variable ****Empathy: Perspective taking**** (the tendency to look at things from another's point of view (Perspective Taking subscale)) has an estimated effect of 0.704. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.70 points on the -9 to 9 scale. The variable ****Empathy: Personal distress**** (the tendency to feel distress in uncomfortable social situations (Personal Distress subscale)) has an estimated effect of 0.115. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.12 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 ingroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 ingroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of 0.129. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.13 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 outgroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 outgroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of -2.666. The results show indicates suggestive but inconclusive evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -2.67 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 control label hidden, N2 ingroup (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 control label hidden, N2 ingroup (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of 0.257. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.26 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 ingroup, N2 outgroup (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 ingroup, N2 outgroup (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of -1.548. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group)

would shift the moral judgment by approximately -1.55 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 outgroup, N2 outgroup (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 outgroup, N2 outgroup (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of -1.194. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.19 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 control label hidden, N2 outgroup (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 control label hidden, N2 outgroup (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of -1.042. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.04 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 ingroup, N2 control label hidden (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 ingroup, N2 control label hidden (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of -0.837. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.84 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 outgroup, N2 control label hidden (ref = N1 control label hidden, N2 control label hidden)**** (the N1-N2 group structure comparing N1 outgroup, N2 control label hidden (ref = N1 control label hidden, N2 control label hidden)) has an estimated effect of -0.568. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.57 points on the -9 to 9 scale. The variable ****Player-victim outgroup**** (whether the player and victim belonged to different groups) has an estimated effect of -0.414. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.41 points on the -9 to 9 scale. The variable ****Engineering participant**** (whether the participant is from an Engineering faculty) has an estimated effect of 0.942. The results show indicates suggestive but inconclusive evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.94 points on the -9 to 9 scale. The variable ****Man participant**** (being a man (relative to being a woman)) has an estimated effect of 0.739. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.74 points on the -9 to 9 scale. The variable ****Age**** (the participant's age) has an estimated effect of 0.349. The results show indicates very strong evidence of an effect, showing that this variable leads to an increase (higher scores) in the moral judgment. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.35 points on the -9 to 9 scale. The variable ****Economic status**** (the socioeconomic contextual stratum of the participant's background) has an estimated effect of 0.338. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable

(or moving from the reference group to this group) would shift the moral judgment by approximately 0.34 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 ingroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) x Player-victim outgroup**** (the N1-N2 group structure comparing N1 ingroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) in combination with whether the player and victim belonged to different groups) has an estimated effect of -1.441. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.44 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 outgroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) x Player-victim outgroup**** (the N1-N2 group structure comparing N1 outgroup, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) in combination with whether the player and victim belonged to different groups) has an estimated effect of 1.869. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 1.87 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 control label hidden, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) x Player-victim outgroup**** (the N1-N2 group structure comparing N1 control label hidden, N2 ingroup (ref = N1 control label hidden, N2 control label hidden) in combination with whether the player and victim belonged to different groups) has an estimated effect of -0.325. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.33 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 ingroup, N2 outgroup (ref = N1 control label hidden, N2 control label hidden) x Player-victim outgroup**** (the N1-N2 group structure comparing N1 ingroup, N2 outgroup (ref = N1 control label hidden, N2 control label hidden) in combination with whether the player and victim belonged to different groups) has an estimated effect of 1.253. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 1.25 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 outgroup, N2 outgroup (ref = N1 control label hidden, N2 control label hidden) x Player-victim outgroup**** (the N1-N2 group structure comparing N1 outgroup, N2 outgroup (ref = N1 control label hidden, N2 control label hidden) in combination with whether the player and victim belonged to different groups) has an estimated effect of 2.518. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 2.52 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 control label hidden, N2 outgroup (ref = N1 control label hidden, N2 control label hidden) x Player-victim outgroup**** (the N1-N2 group structure comparing N1 control label hidden, N2 outgroup (ref = N1 control label hidden, N2 control label hidden) in combination with whether the player and victim belonged to different groups) has an estimated effect of -1.777. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.78 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 ingroup, N2 control label hidden (ref = N1 control label hidden, N2 control label hidden) x Player-victim outgroup**** (the N1-N2 group structure comparing N1 ingroup, N2 control

label hidden (ref = N1 control label hidden, N2 control label hidden) in combination with whether the player and victim belonged to different groups) has an estimated effect of -1.342. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.34 points on the -9 to 9 scale. The variable ****Negotiator-side structure: N1 outgroup, N2 control label hidden (ref = N1 control label hidden, N2 control label hidden) x Player-victim outgroup**** (the N1-N2 group structure comparing N1 outgroup, N2 control label hidden (ref = N1 control label hidden, N2 control label hidden) in combination with whether the player and victim belonged to different groups) has an estimated effect of -1.091. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.09 points on the -9 to 9 scale. The ****Judgment variation (log-scale)**** is 2.328, which indicates the level of dispersion or 'noise' in the latent judgments. This reflects the inherent variety of moral opinions across the sample.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.928$, $p = 1.593e-32$). The primary Tobit model still uses participant-clustered standard errors by id,.

9.3 H3: Empathy x Judged-Status Moderation

This section tests whether empathy slopes differ across judged-negotiator status categories after retaining decision outcome, judged-status x decision terms, counterpart status, and observer-side victim alignment when that predictor is meaningful. As in H1, the victim and bystander subsets use different formulas only where observer-only predictors would otherwise be structurally fixed.

9.3.1 Active Empathy-Construct Specification

Model B subset-specific formulas.

Victim: judgement

```
textasciitilde iri_fs + iri_ec + iri_pt + iri_pd + judged_ingroup + judged_outgroup + decision_accept + decision_accept:judged_ingroup + decision_accept:judged_outgroup + iri_fs:judged_ingroup + iri_ec:judged_ingroup + iri_pt:judged_ingroup + iri_pd:judged_ingroup + iri_fs:judged_outgroup + iri_ec:judged_outgroup + iri_pt:judged_outgroup + iri_pd:judged_outgroup + counterpart_ingroup + counterpart_outgroup + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)
```

Bystander: judgement

```
textasciitilde iri_fs + iri_ec + iri_pt + iri_pd + judged_ingroup + judged_outgroup + decision_accept + decision_accept:judged_ingroup + decision_accept:judged_outgroup + iri_fs:judged_ingroup + iri_ec:judged_ingroup + iri_pt:judged_ingroup + iri_pd:judged_ingroup + iri_fs:judged_outgroup + iri_ec:judged_outgroup + iri_pt:judged_outgroup + iri_pd:judged_outgroup + counterpart_ingroup + counterpart_outgroup + observer_victim_outgroup + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)
```

Victim subset

Table 10: H3 active model: empathy-construct Tobit coefficients (Tobit estimator; Victim subset: 2,430 observations from 243 participants).

Predictor	Estimate	Std. Error	p-value
Base	6.176	3.263	0.058+
FS	0.204	0.611	0.739
EC	0.298	0.781	0.703
PT	-0.023	0.739	0.975
PD	-0.712	0.680	0.295
N1-In	-4.783	2.972	0.108
N1-Out	-3.255	3.022	0.282
Acc	-8.780	0.951	<0.001***
N2-In	-0.160	0.576	0.781
N2-Out	-0.218	0.550	0.693
Eng part.	0.884	0.618	0.153
Man	-0.312	0.666	0.639
Age	0.004	0.127	0.976
SES	0.273	0.293	0.352
Slot 2	-0.082	0.447	0.854
Acc x N1-In	-1.495	1.024	0.144
Acc x N1-Out	-1.199	1.046	0.252
FS x N1-In	0.922	0.764	0.227
EC x N1-In	-0.673	0.960	0.484
PT x N1-In	2.012	0.979	0.040*
PD x N1-In	0.339	0.760	0.655
FS x N1-Out	-0.417	0.779	0.592
EC x N1-Out	-1.039	1.081	0.336
PT x N1-Out	2.000	1.020	0.050+
PD x N1-Out	1.304	0.867	0.133
Log(s)	2.341	0.053	<0.001***

Tobit estimator

Note. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The ****Baseline judgment**** represents the starting moral score when all continuous variables are at their average and all group variables are at their reference levels (for the active negotiator-status terms, the control-labeled condition). In this model, that baseline is estimated at 6.176 ($p=0.058$). The variable ****Empathy: Fantasy scale**** (the tendency to imaginatively get involved in fictitious stories (Fantasy subscale)) has an estimated effect of 0.204. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.20 points on the -9 to 9 scale. The variable ****Empathy: Empathic concern**** (the tendency to feel sympathy and compassion for

others (Empathic Concern subscale)) has an estimated effect of 0.298. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.30 points on the -9 to 9 scale. The variable ****Empathy: Perspective taking**** (the tendency to look at things from another's point of view (Perspective Taking subscale)) has an estimated effect of -0.023. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.02 points on the -9 to 9 scale. The variable ****Empathy: Personal distress**** (the tendency to feel distress in uncomfortable social situations (Personal Distress subscale)) has an estimated effect of -0.712. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.71 points on the -9 to 9 scale. The variable ****N1 ingroup**** (whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of -4.783. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -4.78 points on the -9 to 9 scale. The variable ****N1 outgroup**** (whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -3.255. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -3.25 points on the -9 to 9 scale. The variable ****Harm Accepted**** (whether N1 accepted the harmful deal) has an estimated effect of -8.780. The results show indicates very strong evidence of an effect, showing that this variable leads to a decrease (lower, more severe scores) in the moral judgment. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -8.78 points on the -9 to 9 scale. The variable ****N2 ingroup**** (whether N2 was ingroup rather than in the control-labeled baseline) has an estimated effect of -0.160. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.16 points on the -9 to 9 scale. The variable ****N2 outgroup**** (whether N2 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -0.218. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.22 points on the -9 to 9 scale. The variable ****Engineering participant**** (whether the participant is from an Engineering faculty) has an estimated effect of 0.884. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.88 points on the -9 to 9 scale. The variable ****Man participant**** (being a man (relative to being a woman)) has an estimated effect of -0.312. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.31 points on the -9 to 9 scale. The variable ****Age**** (the participant's age) has an estimated

effect of 0.004. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.00 points on the -9 to 9 scale. The variable ****Economic status**** (the socioeconomic contextual stratum of the participant's background) has an estimated effect of 0.273. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.27 points on the -9 to 9 scale. The variable ****Harm Accepted x N1 ingroup**** (whether N1 accepted the harmful deal in combination with whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of -1.495. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.49 points on the -9 to 9 scale. The variable ****Harm Accepted x N1 outgroup**** (whether N1 accepted the harmful deal in combination with whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -1.199. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.20 points on the -9 to 9 scale. The variable ****Empathy: Fantasy scale x N1 ingroup**** (the tendency to imaginatively get involved in fictitious stories (Fantasy subscale) in combination with whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of 0.922. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.92 points on the -9 to 9 scale. The variable ****Empathy: Empathic concern x N1 ingroup**** (the tendency to feel sympathy and compassion for others (Empathic Concern subscale) in combination with whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of -0.673. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.67 points on the -9 to 9 scale. The variable ****Empathy: Perspective taking x N1 ingroup**** (the tendency to look at things from another's point of view (Perspective Taking subscale) in combination with whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of 2.012. The results show indicates clear evidence of an effect, showing that this variable leads to an increase (higher scores) in the moral judgment. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 2.01 points on the -9 to 9 scale. Because this is a continuous-by-discrete interaction, the positive coefficient indicates that the effect of empathy on the judgment is strengthened (pushed higher) for this specific condition compared to the baseline. This means the 'empathy gap' in judgment is less severe here. The variable ****Empathy: Personal distress x N1 ingroup**** (the tendency to feel distress in uncomfortable social situations (Personal Distress subscale) in combination with whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of 0.339. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.34 points on the -9 to 9 scale. The variable ****Empathy: Fantasy scale x N1 outgroup**** (the tendency to imaginatively get involved in fictitious

stories (Fantasy subscale) in combination with whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -0.417. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.42 points on the -9 to 9 scale. The variable ****Empathy: Empathic concern x N1 outgroup**** (the tendency to feel sympathy and compassion for others (Empathic Concern subscale) in combination with whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -1.039. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.04 points on the -9 to 9 scale. The variable ****Empathy: Perspective taking x N1 outgroup**** (the tendency to look at things from another's point of view (Perspective Taking subscale) in combination with whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of 2.000. The results show indicates suggestive but inconclusive evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 2.00 points on the -9 to 9 scale. Because this is a continuous-by-discrete interaction, the positive coefficient indicates that the effect of empathy on the judgment is strengthened (pushed higher) for this specific condition compared to the baseline. This means the 'empathy gap' in judgment is less severe here. The variable ****Empathy: Personal distress x N1 outgroup**** (the tendency to feel distress in uncomfortable social situations (Personal Distress subscale) in combination with whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of 1.304. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 1.30 points on the -9 to 9 scale. The ****Judgment variation (log-scale)**** is 2.341, which indicates the level of dispersion or 'noise' in the latent judgments. This reflects the inherent variety of moral opinions across the sample.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.962$, $p = 8.649e-25$). The primary Tobit model still uses participant-clustered standard errors by id,.

Bystander subset

Table 11: H3 active model: empathy-construct Tobit coefficients (Tobit estimator; Bystander subset: 2,430 observations from 243 participants).

Predictor	Estimate	Std. Error	p-value
Base	-4.375	3.171	0.168
FS	0.947	0.633	0.135
EC	-0.468	0.862	0.587
PT	1.416	0.757	0.061+
PD	0.259	0.671	0.700
N1-In	5.475	2.674	0.041*
N1-Out	5.964	2.510	0.017*
Acc	-9.032	0.945	<0.001***
N2-In	0.140	0.455	0.759
N2-Out	-0.152	0.479	0.751
V-Out	-0.108	0.431	0.802
Eng part.	0.992	0.540	0.066+
Man	0.335	0.596	0.574
Age	0.290	0.102	0.004**
SES	0.236	0.273	0.387
Slot 2	0.587	0.427	0.169
Acc x N1-In	0.694	1.061	0.513
Acc x N1-Out	0.383	1.101	0.728
FS x N1-In	-0.642	0.741	0.387
EC x N1-In	-0.018	0.913	0.984
PT x N1-In	-1.399	0.878	0.111
PD x N1-In	-0.776	0.724	0.284
FS x N1-Out	-1.026	0.674	0.128
EC x N1-Out	-1.122	0.974	0.249
PT x N1-Out	-0.771	0.851	0.365
PD x N1-Out	0.316	0.781	0.685
Log(s)	2.219	0.054	<0.001***

Tobit estimator

Note. Abbrev.: FS / EC / PT / PD = IRI subscales; N1 = judged negotiator; N2 = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The **Baseline judgment** represents the starting moral score when all continuous variables are at their average and all group variables are at their reference levels (for the active negotiator-status terms, the control-labeled condition). In this model, that baseline is estimated at -4.375 (p=0.168). The variable **Empathy: Fantasy scale** (the tendency to imaginatively get involved in fictitious stories (Fantasy subscale)) has an estimated effect of 0.947. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.95 points on the -9 to 9 scale. The

variable ****Empathy: Empathic concern**** (the tendency to feel sympathy and compassion for others (Empathic Concern subscale)) has an estimated effect of -0.468. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.47 points on the -9 to 9 scale. The variable ****Empathy: Perspective taking**** (the tendency to look at things from another's point of view (Perspective Taking subscale)) has an estimated effect of 1.416. The results show indicates suggestive but inconclusive evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 1.42 points on the -9 to 9 scale. The variable ****Empathy: Personal distress**** (the tendency to feel distress in uncomfortable social situations (Personal Distress subscale)) has an estimated effect of 0.259. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.26 points on the -9 to 9 scale. The variable ****N1 ingroup**** (whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of 5.475. The results show indicates clear evidence of an effect, showing that this variable leads to an increase (higher scores) in the moral judgment. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 5.48 points on the -9 to 9 scale. The variable ****N1 outgroup**** (whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of 5.964. The results show indicates clear evidence of an effect, showing that this variable leads to an increase (higher scores) in the moral judgment. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 5.96 points on the -9 to 9 scale. The variable ****Harm Accepted**** (whether N1 accepted the harmful deal) has an estimated effect of -9.032. The results show indicates very strong evidence of an effect, showing that this variable leads to a decrease (lower, more severe scores) in the moral judgment. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -9.03 points on the -9 to 9 scale. The variable ****N2 ingroup**** (whether N2 was ingroup rather than in the control-labeled baseline) has an estimated effect of 0.140. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.14 points on the -9 to 9 scale. The variable ****N2 outgroup**** (whether N2 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -0.152. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.15 points on the -9 to 9 scale. The variable ****Observer-side victim outgroup**** (whether the victim belonged to a different faculty than the observer (Outgroup)) has an estimated effect of -0.108. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.11 points on the -9 to 9 scale. The variable ****Engineering participant**** (whether the participant is from an Engineering faculty) has an estimated effect of 0.992. The results show indicates suggestive but inconclusive evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would

shift the moral judgment by approximately 0.99 points on the -9 to 9 scale. The variable ****Man participant**** (being a man (relative to being a woman)) has an estimated effect of 0.335. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.33 points on the -9 to 9 scale. The variable ****Age**** (the participant's age) has an estimated effect of 0.290. The results show indicates strong evidence of an effect, showing that this variable leads to an increase (higher scores) in the moral judgment. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.29 points on the -9 to 9 scale. The variable ****Economic status**** (the socioeconomic contextual stratum of the participant's background) has an estimated effect of 0.236. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.24 points on the -9 to 9 scale. The variable ****Harm Accepted x N1 ingroup**** (whether N1 accepted the harmful deal in combination with whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of 0.694. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.69 points on the -9 to 9 scale. The variable ****Harm Accepted x N1 outgroup**** (whether N1 accepted the harmful deal in combination with whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of 0.383. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.38 points on the -9 to 9 scale. The variable ****Empathy: Fantasy scale x N1 ingroup**** (the tendency to imaginatively get involved in fictitious stories (Fantasy subscale) in combination with whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of -0.642. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.64 points on the -9 to 9 scale. The variable ****Empathy: Empathic concern x N1 ingroup**** (the tendency to feel sympathy and compassion for others (Empathic Concern subscale) in combination with whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of -0.018. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.02 points on the -9 to 9 scale. The variable ****Empathy: Perspective taking x N1 ingroup**** (the tendency to look at things from another's point of view (Perspective Taking subscale) in combination with whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of -1.399. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.40 points on the -9 to 9 scale. The variable ****Empathy: Personal distress x N1 ingroup**** (the tendency to feel distress in uncomfortable social situations (Personal Distress subscale) in combination with whether N1 was ingroup rather than in the control-labeled baseline) has an estimated effect of -0.776. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable.

For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.78 points on the -9 to 9 scale. The variable ****Empathy: Fantasy scale x N1 outgroup**** (the tendency to imaginatively get involved in fictitious stories (Fantasy subscale) in combination with whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -1.026. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.03 points on the -9 to 9 scale. The variable ****Empathy: Empathic concern x N1 outgroup**** (the tendency to feel sympathy and compassion for others (Empathic Concern subscale) in combination with whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -1.122. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -1.12 points on the -9 to 9 scale. The variable ****Empathy: Perspective taking x N1 outgroup**** (the tendency to look at things from another's point of view (Perspective Taking subscale) in combination with whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of -0.771. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately -0.77 points on the -9 to 9 scale. The variable ****Empathy: Personal distress x N1 outgroup**** (the tendency to feel distress in uncomfortable social situations (Personal Distress subscale) in combination with whether N1 belonged to a different faculty than the reference participant (Outgroup)) has an estimated effect of 0.316. The results show indicates little to no evidence of an effect, meaning the current data doesn't reliably show a relationship for this specific variable. For example, a 1-unit increase in this variable (or moving from the reference group to this group) would shift the moral judgment by approximately 0.32 points on the -9 to 9 scale. The ****Judgment variation (log-scale)**** is 2.219, which indicates the level of dispersion or 'noise' in the latent judgments. This reflects the inherent variety of moral opinions across the sample.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.971$, $p = 7.124e-22$). The primary Tobit model still uses participant-clustered standard errors by id,.

10 Discussion and Limitations

While the dual-estimator strategy strengthens the analysis, several limitations remain. First, both inference strategies assume independence between participants, which may still be violated by broader shared institutional or contextual shocks. Second, we assume a normal distribution for the latent errors ϵ_{ij} in the Tobit branch; departures from normality could affect the consistency of the maximum likelihood estimators. Finally, the use of unstandardized averages for empathy assumes a linear relationship between the survey score and actual moral preference.

11 Conclusion

Based on the interval-censored Tobit estimations using the four empathy constructs, empathy and relational judgment structure have been documented together under Option 2 through negotiator-

level relational predictors rather than descriptive case labels.